

Integrals – 7

Computing an integral without an antiderivative

Homework

11 сентября 2026 года

Theoretical minimum. The checklist of the class sheet: symmetries of the interval and the king's rule; functional equations solved by a substitution of finite order, in particular the order three map $x \mapsto 1 - \frac{1}{x}$; a parameter inserted so that the derivative in it is elementary; a power series recognised in closed form and integrated term by term; the beta integral $\int_0^1 x^p(1-x)^q dx = \frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+2)}$ and the fact that $\frac{(2n+1)!}{(n!)^2} (x(1-x))^n$ is an approximate identity concentrating at $x = \frac{1}{2}$; the location of the mass of $\int x^r g(x) dx$ for large r . A region in the plane may also be cut along the curve where two expressions under a maximum change places.

1. (Putnam 1987) Evaluate $\int_2^4 \frac{\sqrt{\ln(9-x)} dx}{\sqrt{\ln(9-x)} + \sqrt{\ln(x+3)}}$.
2. (Putnam 2016) Suppose that f is a function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) + f\left(1 - \frac{1}{x}\right) = \arctan x$$

for all real $x \neq 0$. Find $\int_0^1 f(x) dx$.

3. (Putnam 1985) Let $I_m = \int_0^{2\pi} \cos(x) \cos(2x) \dots \cos(mx) dx$. For which integers m , $1 \leq m \leq 10$, is $I_m \neq 0$?
4. (Putnam 1989) Evaluate $\int_0^a \int_0^b e^{\max\{b^2 x^2, a^2 y^2\}} dy dx$, where a and b are positive.
5. (Putnam 2005) Evaluate $\int_0^1 \frac{\ln(x+1)}{x^2+1} dx$.

6. (Putnam 1997) Evaluate

$$\int_0^\infty \left(x - \frac{x^3}{2} + \frac{x^5}{2 \cdot 4} - \frac{x^7}{2 \cdot 4 \cdot 6} + \dots \right) \left(1 + \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} + \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \right) dx.$$

7. (Putnam 1992) Define $C(\alpha)$ to be the coefficient of x^{1992} in the power series expansion about $x = 0$ of $(1+x)^\alpha$. Evaluate

$$\int_0^1 C(-y-1) \left(\frac{1}{y+1} + \frac{1}{y+2} + \frac{1}{y+3} + \dots + \frac{1}{y+1992} \right) dy.$$

8. (Putnam 2008) Let $F_0(x) = \ln x$. For $n \geq 0$ and $x > 0$, let $F_{n+1}(x) = \int_0^x F_n(t) dt$. Evaluate $\lim_{n \rightarrow \infty} \frac{n! F_n(1)}{\ln n}$.

9. (SEEMOUS 2009)

- a) Calculate the limits $\lim_{n \rightarrow \infty} \frac{(2n+1)!}{(n!)^2} \int_0^1 (x(1-x))^n x^k dx$, where $k \in \mathbb{N}$.
- b) Calculate the limit $\lim_{n \rightarrow \infty} \frac{(2n+1)!}{(n!)^2} \int_0^1 (x(1-x))^n f(x) dx$, where $f: [0, 1] \rightarrow \mathbb{R}$ is a continuous function.

10. (Putnam 2011) Find a real number c and a positive number L for which

$$\lim_{r \rightarrow \infty} r^c \frac{\int_0^{\pi/2} x^r \sin x \, dx}{\int_0^{\pi/2} x^r \cos x \, dx} = L.$$